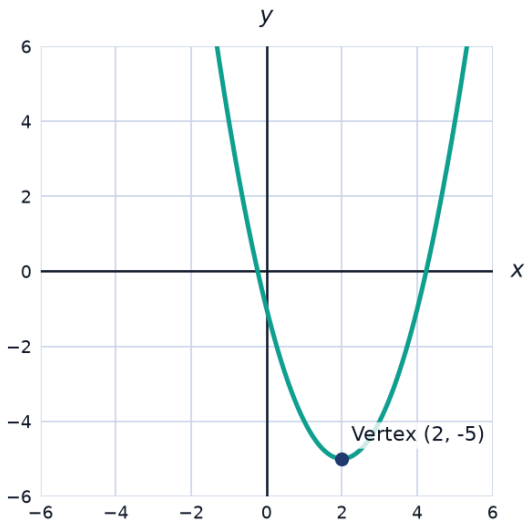
Number & Algebra

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solve by factorisation**

Consider the quadratic function $f(x) = x^2 + bx + c$. The graph of $f(x)$ is shown below. The vertex of the parabola is at $(2, -5)$ and it passes through the point $(0, -1)$. Determine the values of b and c.

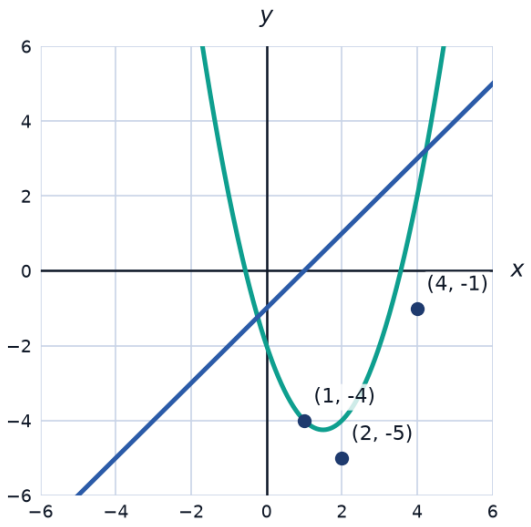


2 **[Foundational]** • **Target: Solve quadratics by completing the square**

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of k.

3 **[Foundational]** • **Target: Solve using the quadratic formula**

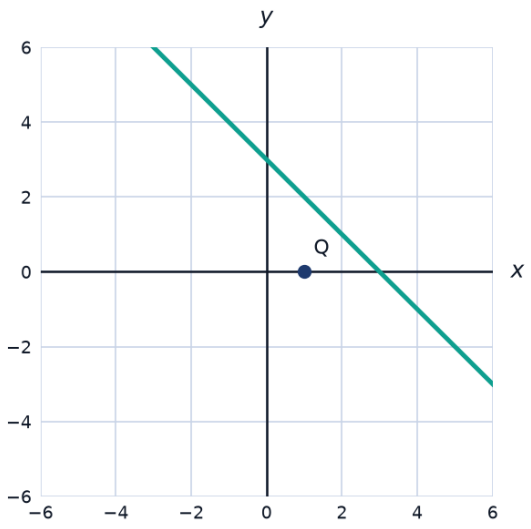
The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of a.



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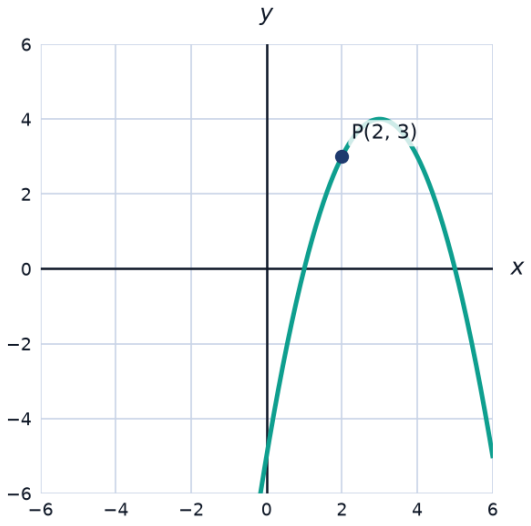
4 [Foundational] • Target: Discriminant of a quadratic

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.



5 [Foundational] • Target: Solve quadratics using various methods

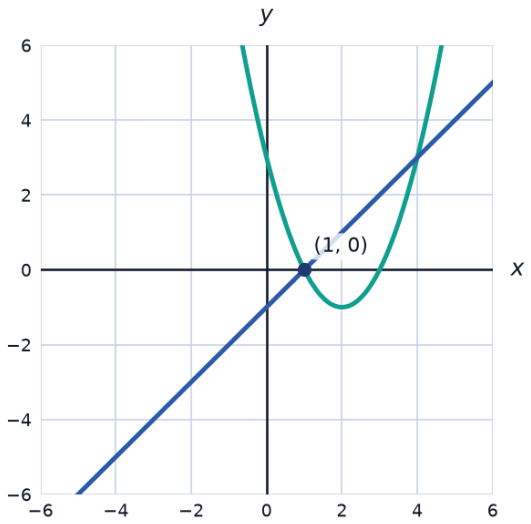
Consider the function $f(x) = -x^2 + 6x - 5$. The graph of this function is shown below. A tangent line to the graph at point $P(2, 3)$ intersects the x-axis at point Q. Determine the coordinates of point Q.



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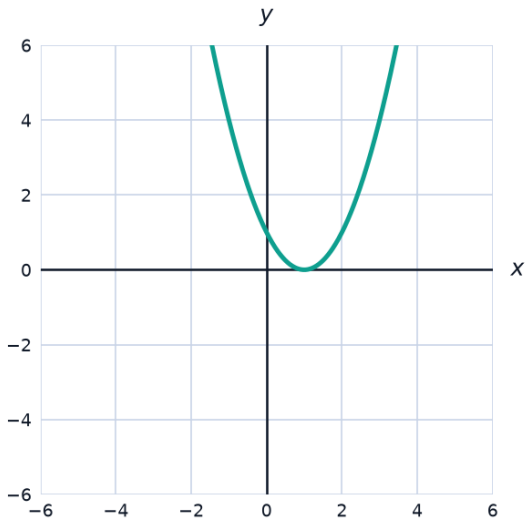
6 [Foundational] • Target: Graph basic quadratics

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.



7 [Foundational] • Target: Solve by factorisation

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.

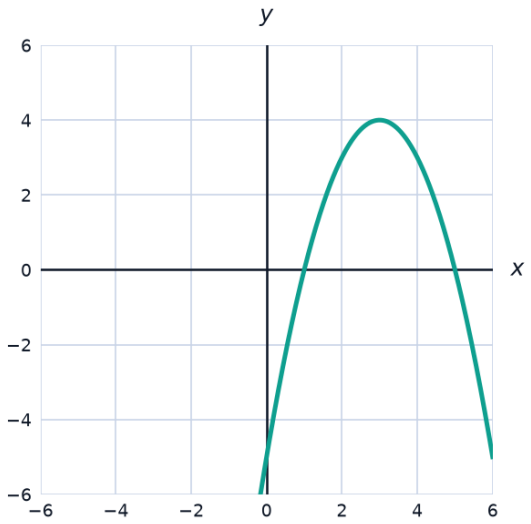


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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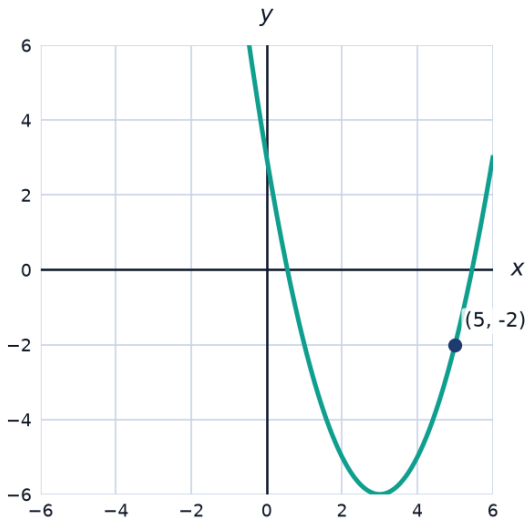
8 **[Intermediate]** • **Target: Solve quadratics by completing the square**

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.



9 **[Intermediate]** • **Target: Solve using the quadratic formula**

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(5, -2)$. Find the value of k.



10 **[Intermediate]** • **Target: Discriminant of a quadratic**

Consider the function $f(x) = e^{-x^2}$ and the line $y = k$. The graph of $y = f(x)$ is concave up for $x \in (-\infty, -a) \cup (a, \infty)$ and concave down for $x \in (-a, a)$. Find the exact value of a^2.

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11 [Intermediate] • Target: Solve quadratics using various methods

The depth of water $h(t)$ in meters at a harbor entrance t hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

12 [Intermediate] • Target: Graph basic quadratics

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the exact value of the sum of the x-coordinates of the local maximum and local minimum points of the graph of $y = f(x)$.

13 [Intermediate] • Target: Solve by factorisation

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $x \neq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

14 [Intermediate] • Target: Solve quadratics by completing the square

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 6.01 AI SL — Specific Topics

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

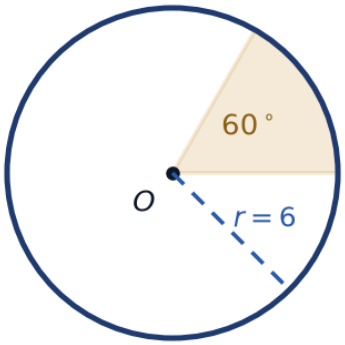
16 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 2.01 Functions

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2+9}}\,dx$ using the substitution $u = x^2 + 9$.

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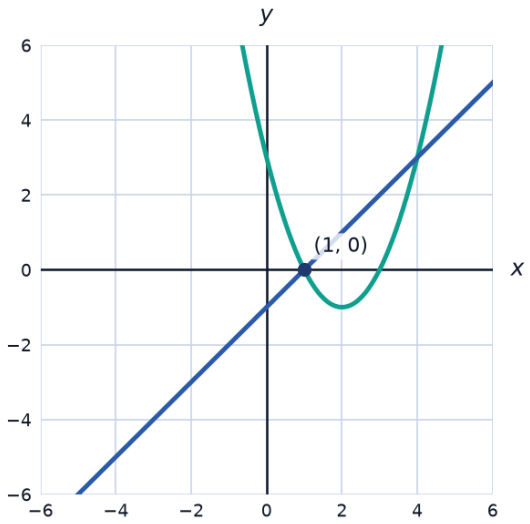
17 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 3.01 Geometry & Trigonometry

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.



18 [Challenging] • Target: Graph basic quadratics • Cross-Lesson: 4.01 Statistics & Probability

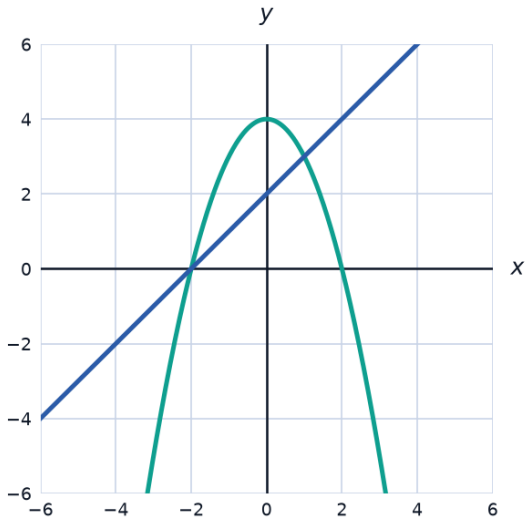
Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



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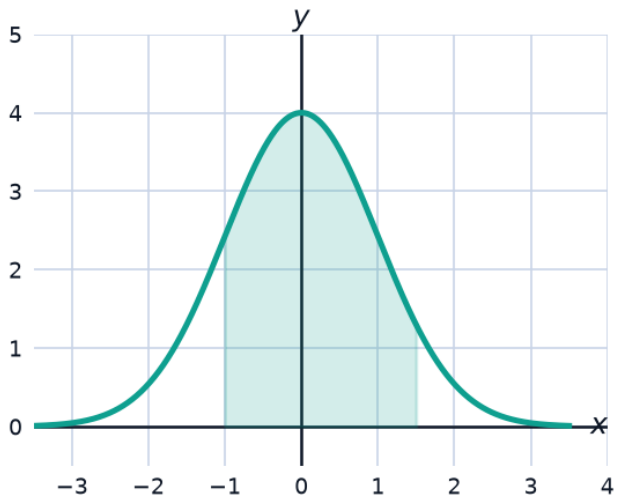
19 [Challenging] • Target: Solve by factorisation • Cross-Lesson: 5.01 Calculus

Solve the logarithmic equation $\log_3(x+5) + \log_3(x-1) = 2 + \log_3(2)$ for all real values of x.



20 [Challenging] • Target: Solve quadratics by completing the square • Cross-Lesson: 6.01 AI SL — Specific Topics

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

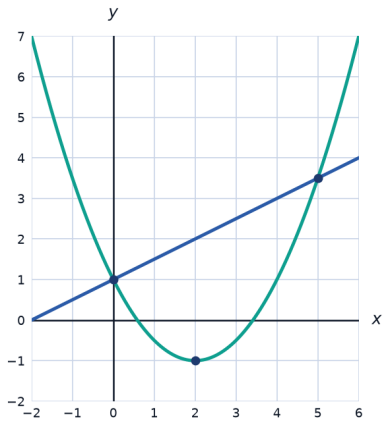


Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Solve by factorisation

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(2)$.



a. 7

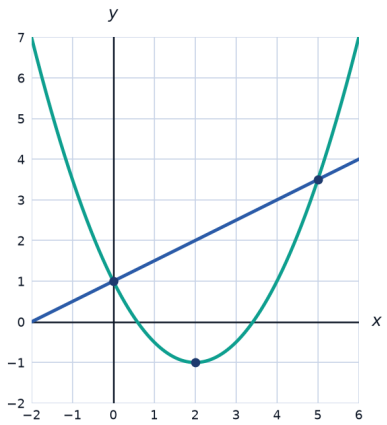
b. 9

c. 4

d. 11

22 [Foundational] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

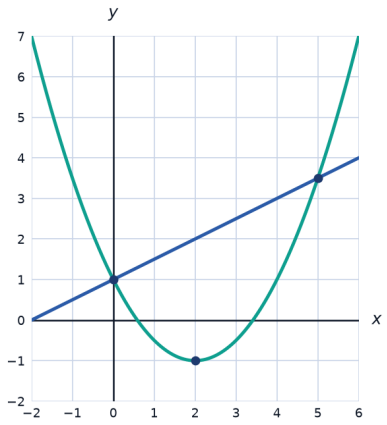
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Solve by factorisation

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(5)$.



a. 13

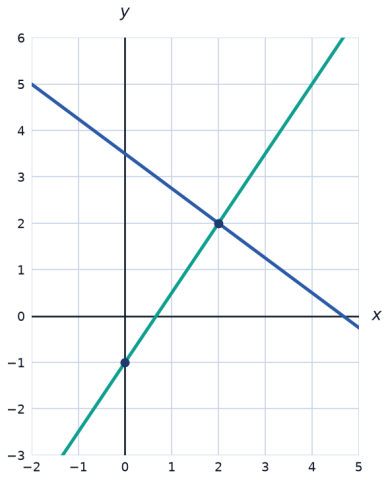
b. 15

c. 10

d. 17

24 [Foundational] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(2)$.



a. 10

b. 13

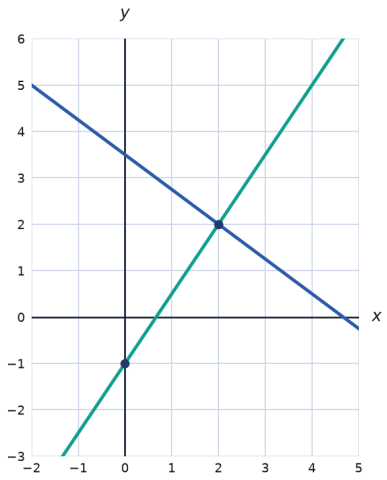
c. 6

d. 14

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25 [Foundational] • Target: Solve quadratics by completing the square

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

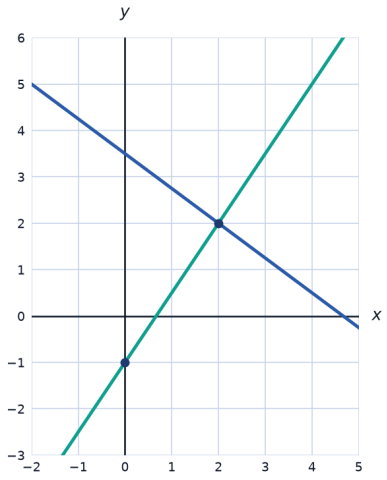
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(5)$.



a. 19

b. 22

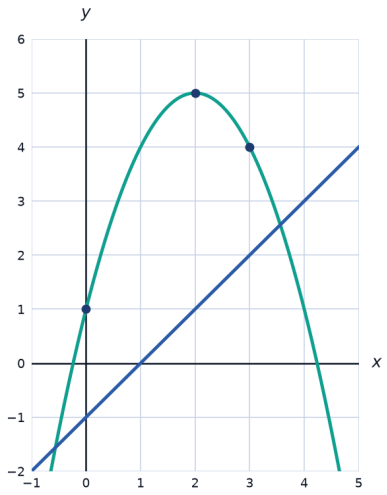
c. 15

d. 23

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27 [Foundational] • Target: Solve using the quadratic formula

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(2)$.



a. 13

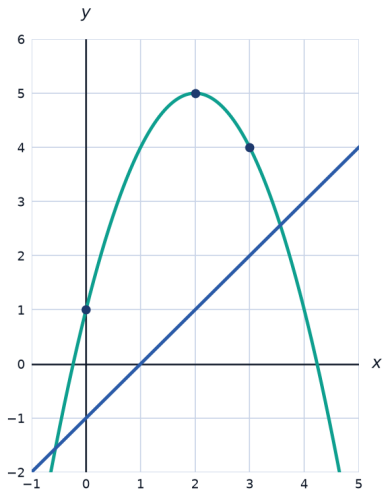
b. 17

c. 8

d. 17

28 [Foundational] • Target: Solve using the quadratic formula

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Solve by factorisation

A data set recorded for solve by factorisation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

31 [Intermediate] • Target: Solve by factorisation

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

32 [Intermediate] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Solve quadratics by completing the square

A data set recorded for solve quadratics by completing the square consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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34 [Intermediate] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

35 [Intermediate] • Target: Solve quadratics by completing the square

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

36 [Intermediate] • Target: Solve using the quadratic formula

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

37 [Intermediate] • Target: Solve using the quadratic formula

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

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39 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 5.01 Calculus

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 6.01 AI SL — Specific Topics

A data set recorded for discriminant of a quadratic consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

41 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 2.01 Functions

For the model $f(x) = 5x + 6$ in discriminant of a quadratic, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

42 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

43 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for solve quadratics using various methods consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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44 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 6x + 7$ in solve quadratics using various methods, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

45 [Challenging] • Target: Graph basic quadratics • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 2x + 8$ in graph basic quadratics, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

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